

Anywhere Hoop: Virtual Free Throw Training System

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ABSTRACT

To complete successfully a high percentage of free throws in basketball, the shooter must achieve a stable trajectory with the ball. A player must practice shooting repeatedly to shoot a ball with a stable trajectory. However, traditional practice methods require the preparation of a real basketball hoop, which has made it difficult for some players to prepare a practice environment. We propose a training method for free throws using a virtual basketball hoop. In this paper, we present an implementation of the proposed method and experimental results showing its effectiveness.

CCS CONCEPTS

• **Human-centered computing** → *Mixed / augmented reality.*

KEYWORDS

Basketball, Augmented reality, Head-mounted display

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1 INTRODUCTION

Basketball is a popular worldwide sport, with an estimated 450 million players [2]. Basketball incorporates a shot called a free throw in which an unhindered shot is attempted from behind a set line after a foul is committed on the shooter by a member of the opposing team. A free throw represents a very important scoring opportunity and is said to have a significant impact on the outcome of a game [1].

The key to completing successfully a high percentage of free throws is to develop a stable trajectory, i.e., a shooting form such that the ball follows an angle of incidence of 45 degrees [4]. Methods for practicing the shooting form have been proposed to provide feedback on the shooting trajectory during practice [3, 5]. For example, there are commercial applications such as HomeCourt, which uses a smartphone camera to track the movement of a basketball and display its trajectory [3]. It saves video and provides results analyzing whether a shot was successful or not, so that they can be checked after practice. Lin et al. proposed a method that provides

visual feedback through 3D visualization of the shooting trajectory of the player using a head-mounted display (HMD) [5]. To enable this method, they placed multiple cameras on an actual hoop and estimated the trajectory of the ball moving in space.

Unfortunately, the methods in [3] and [5] require a real basketball hoop. A real basketball hoop is too large to prepare in a home. A player could set up a basketball hoop in a large area such as a warehouse every time they wish to practice, but this takes time and requires the preparation of a separate space for the basketball hoop. On the other hand, a public facility can be used as a practice environment. However, it may be difficult to secure a stable practice environment because there may be no such facilities within the player living area, or there may be restrictions on the duration and timing of use. Even if public facilities are stably available, the use of public facilities may be suspended due to an outbreak of infectious diseases such as SARS-CoV-2(COVID-19).

We propose a training method for free throws that provides a virtual basketball hoop and shot trajectory analytics using an HMD and depth-sensing camera. The proposed method mainly comprises of two steps: 1) The user can see a virtual basketball hoop and the ideal shooting trajectory when wearing the HMD. 2) When the user releases the ball, the depth sensing camera senses the trajectory and the HMD displays visual feedback relating to the angle of incidence of the shot. The proposed method makes it possible to practice free throws without using an actual basketball hoop, thus making it easier to secure a practice environment. Another advantage of using a virtual basketball hoop is the ability to add visual aids, such as constantly displaying the ideal ball trajectory. However, there has been no verification of the effectiveness of using a virtual basketball hoop in practicing free throws. The purpose of this paper is to show the training effects that can be expected in free-throw practice with a virtual basketball hoop on a HMD.

2 IMPLEMENTATION

- Implementation details.

We implemented the proposed method, as Figure 1a shows. The proposed method comprises the following steps: 1) After launching the application on the user HMD, which is initiated by reading the QR code as shown in Figure 1a, a virtual basketball hoop and the trajectory of the ball drawn at a 45 degree angle of incidence is displayed, as Figure 1b shows. 2) From the time that the ball enters the frame of the depth sensing camera until it exits the frame, the center coordinates of the ball and the depth information are recorded. The center coordinates are obtained using blob detection. The depth information uses the value of the center coordinates. 3) The linear trajectory of the ball is calculated from the recorded values using the least-squares method. 4) The angle of incidence of the ball is calculated by calculating the angle that the linear trajectory of the ball follows in the horizontal direction. 5) The user HMD displays

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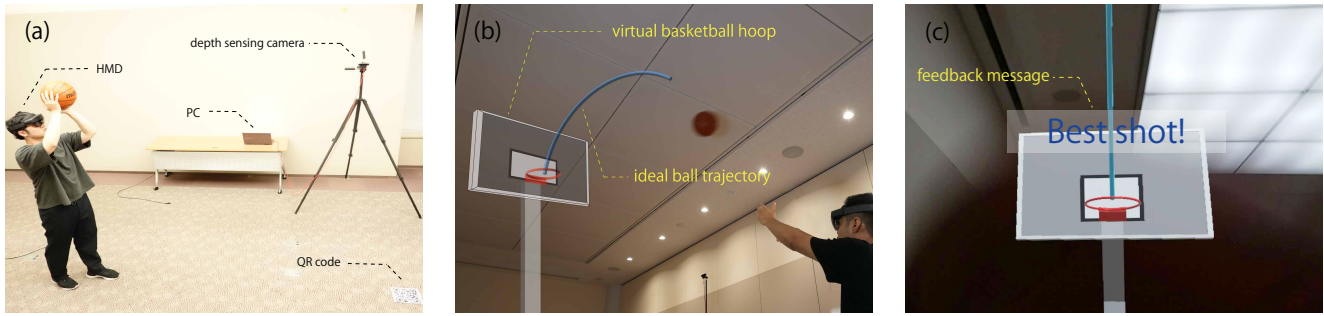


Figure 1: (a) Main devices composing to the proposed system, (b) Virtual basketball hoop displayed on HMD, (c) Feedback message displayed on HMD.

feedback messages according to the estimated angle of incidence, as Figure 1c shows. Different feedback messages, "Little Lower", "Best shot!", and "Little higher!", are displayed for 30 degrees or less, 30 to 50 degrees, and 50 degrees or more, respectively.

- Evaluation setting.

To evaluate the proposed method, we conducted experiments comprising four steps: 1) We explain the experiment and the basketball shooting form to the participant. 2) We have the participant perform 10 shots on an actual basketball hoop. 3) We have the participant perform 50 shots on the virtual basketball hoop. 4) We have the participant once again perform 10 shots on an actual basketball hoop. We evaluated this experiment based on the distance the ball deviates from the center point of the hoop, δ , and the angle of incidence when the ball falls to the height of the basketball hoop, θ . A smaller δ value is better, and an angle closer to 45 degrees for θ is a better result. Both indicators are calculated from the video of the shot in steps 2 and 4.

We conducted the experiments with the cooperation of four men with the average age of 29.75 years. The basketball experience of the men was limited to playing basketball in physical education classes in school. They were informed of the experiment in advance and their consent was obtained before the experiment was conducted.

- Evaluation results.

Figures 2a and 2b show that overall, the average and standard deviation of δ decreased. We also performed Welch's t-test on δ and rejected the null hypothesis that "there is no significant difference in the average of δ before and after practice" at the 5% significance level (p -value = 0.0118). This suggests that the proposed method is effective in developing a sense of distance for free throws. Figure 2c and 2d show the evaluation results of θ . The results in Figure 2c show that the average of θ did not approach 45 degrees. However, Figure 2d shows that the standard deviation decreased for all participants. This suggests that although the ball trajectory is not ideal, the proposed method seems effective from the aspect of stabilizing the ball trajectory.

When participants were interviewed after the experiment regarding the proposed method, all participants indicated that they thought practice using the proposed method was effective. This suggests that the proposed method can contribute to free-throw training. On the other hand, there were also comments such as, "It was difficult to tell if the shot was in or not, and there were situations during practice where I wondered if the ball trajectory

was correct." It is possible that the feedback messages may have had a negative impact on the motivation of some participants to practice.

The proposed method has two limitations. First, the low frame rate (frames per second (FPS)) of the depth sensing camera makes it impossible to acquire accurately depth information for a ball moving quickly through space. This problem could be resolved using a depth sensing camera with a high FPS. Second, it is difficult to experience whether or not the shot made it through the hoop. For an actual basketball hoop, the ball can be judged by the way it bounces and the sound it makes, but in the proposed method, only the displayed feedback message changes. One way to improve, this is to introduce, for example, additional sound feedback and net swishing effects.

3 CONCLUSION

We proposed a free throw training method using a virtual basketball hoop. Practice using the proposed method was shown to improve free-throw performance in terms of developing a sense of distance and trajectory stability. We explained and discussed the mechanism, limitations, and future work for the proposed method.

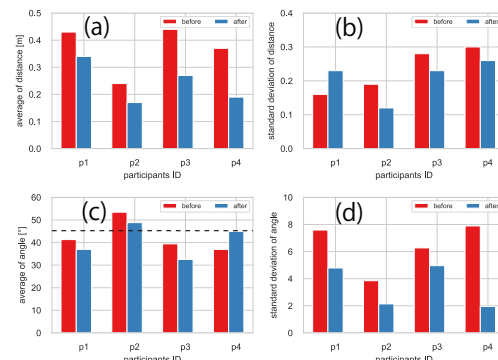


Figure 2: All figures are tabulated from each participant's before and after training results. (a) Average of δ , (b) Standard deviation of δ , (c) Average of θ , (d) Standard deviation of θ .

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